

MCAS PROJECT UPDATE

MedMasters Course Architecture System™

Current Architecture, Decisions, and Next Development Priorities

Project Vision

The MedMasters Course Architecture System™ (MCAS) is a faculty-centered Course Architecture & Compliance Engine designed to help educators build high-quality courses efficiently while maintaining instructional quality, accessibility, compliance, and student engagement.

MCAS is NOT an AI course generator.

MCAS is a structured course design system that uses AI to accelerate and document course development.

Primary Goals:

- Reduce faculty development time
- Improve course quality
- Improve consistency
- Support accessibility compliance
- Support RSI compliance
- Support accreditation and quality assurance processes
- Generate reusable course assets

Key Message:

Great Courses Aren't Generated. They're Architected.

Core System Architecture

The approved workflow is:

DISCOVER ↓ DESIGN ↓ ASSET CAPTURE ↓ SCHEDULE & PACING ↓ FRAME ↓ BUILD ↓ ASSESS ↓ ENGAGE
↓ COMPLIANCE ↓ LAUNCH ↓ SCALE

Major System Inputs

The system currently operates using three primary inputs.

Input 1: Faculty Brand Sheet

One Brand Sheet per faculty member.

Purpose:

Defines how the course should feel.

Includes:

- Teaching philosophy
- Faculty voice
- Communication style
- Visual preferences
- Accessibility preferences
- Assessment preferences
- AI collaboration rules
- Student population
- Course design philosophy

The Brand Sheet already exists as an HTML-based intake system.

This should remain independent from the course workbook.

The same Brand Sheet may be used across multiple courses.

Brand Sheet answers:

"What should this course feel like?"

Input 2: MCAS Master Course Workbook

One workbook per course.

Purpose:

Defines what should be built.

The workbook serves as the single source of truth for course architecture.

Current Workbook Structure:

1. Course Setup
2. Faculty Brand Summary
3. Learning Outcomes
4. Asset Library
5. Schedule & Pacing
6. Module Builder
7. Assessments
8. Engagement
9. Lab Components
10. Accessibility
11. RSI
12. Quality Assurance
13. Launch Checklist

The workbook should continue to evolve into a complete course operating system.

Workbook answers:

"What should be built?"

Input 3: Prototype Selection

One prototype selected per course.

Purpose:

Determines how the course functions.

Prototype answers:

"How should the course be organized?"

Current Prototype List:

1. Traditional Online
 2. Hybrid
 3. Team-Based Learning (TBL)
 4. Case-Based
 5. Accelerated
 6. Competency-Based
 7. Course Redesign / Program Review
-

Engine Logic

The engine should function as follows:

Faculty Brand Sheet

+ Course Workbook

+ Course Prototype ↓ MCAS Engine ↓ Course Package

The engine should interpret:

Brand Sheet → Defines voice, tone, visual style, communication style

Workbook → Defines course structure, assets, schedule, outcomes, assessments

Prototype → Defines instructional architecture

Compliance Framework

MCAS includes a dedicated Compliance Phase.

This is a major differentiator.

Accessibility Compliance

Standard:

WCAG 2.2 Level AA

Review Areas:

- Heading structure
- Alt text
- Color contrast
- Accessible tables
- Captioning
- Transcripts
- Accessible documents
- Keyboard navigation

Outputs:

Accessibility Compliance Report

Accessibility Readiness Score

Remediation Recommendations

RSI Compliance

Standard:

Regular and Substantive Interaction

Review Areas:

- Instructor presence
- Direct instruction
- Guided facilitation
- Feedback schedule
- Communication plan
- Student support

Outputs:

RSI Compliance Report

RSI Evidence Matrix

RSI Readiness Score

Quality Assurance

Review Areas:

- Alignment
- Navigation
- Consistency
- Student experience
- Outcome coverage

Outputs:

Quality Assurance Report

Course Readiness Score

Workbook Evolution Plan

Future Workbook Features:

- Dashboard
- Progress tracker
- Completion percentage
- WCAG score
- RSI score
- Readiness score
- Automated calculations
- Faculty workload analysis
- Student workload analysis
- Compliance dashboard

The workbook should become the primary operating document for each course.

MCAS Outputs

The engine should eventually generate:

Course Blueprint

- Course vision
 - Outcome alignment
 - Module architecture
 - Assessment architecture
-

Course Content

- Home pages
 - Module pages
 - Weekly checklists
 - Assignments
 - Discussions
 - Learning activities
-

Assessments

- Quizzes

- Exams
 - Rubrics
 - Projects
 - Authentic assessments
-

Engagement Assets

- Team-Based Learning activities
 - Clinical cases
 - Simulations
 - Draw-to-Know-It activities
 - Reflection activities
-

Compliance Reports

- Accessibility Report
 - RSI Report
 - Quality Assurance Report
-

Certification Package

- Course Blueprint Report
 - Accessibility Compliance Report
 - RSI Compliance Report
 - Quality Assurance Report
 - Launch Checklist
 - Course Readiness Certification
-

Branding Guidelines

MCAS should match existing MedMasters branding.

Typography:

- Plus Jakarta Sans

Primary Colors:

Navy

0B1530

Rust

8B3A2E

Gold

C9A14A

Cream

F5F1E8

Dark Background

060A18

Design Style:

- Minimalist
 - Premium
 - Academic
 - Consulting-quality
 - Large whitespace
 - Clean sectioning
 - Numbered process architecture
-

Current Development Status

Completed:

✓ Brand Sheet Architecture ✓ Workbook Architecture ✓ Compliance Framework ✓ MCAS Workflow ✓
Prototype Framework ✓ Certification Concept

In Progress:

☐ Workbook Version 3 ☐ Dashboard Design ☐ Prototype Specifications ☐ Accessibility Engine ☐ RSI
Engine

Next Priority:

Develop detailed Prototype Specifications for each course model.

This is now the most important remaining design component because it determines exactly how each course package will be generated.

The MCAS system should continue evolving toward a complete Course Architecture & Compliance Engine rather than a simple content generator.